Waipapa Taumata Rau University of Auckland

COMPSCI 713 - AI Fundamentals

Examines the core concepts and techniques in AI, including breakthroughs in symbolic AI, machine learning, and neural networks. Real-world applications are presented, with a focus on AI research in Aotearoa/NZ and ethical considerations. The course is designed to be accessible to students with limited programming experience.

Course Details

Course Code	COMPSCI 713
Course Title	AI Fundamentals
Course Department	Computer Science
Course Level	Stage 7 course
Points	15
Delivery Mode	Campus Experience

Course Offerings

The information below indicates when the course is normally offered. Availability may change up until semester start.

Campus	Semester Usually Offered
City	Semester One

Course Overview

Examines the core concepts and techniques in AI. Students will be exposed to the various schools of thought that have shaped AI since its inception in the 1940s and the relation of these ideas to prior work. This will include an introduction to pioneering breakthroughs in symbolic AI, including heuristic search, constraint satisfaction, and knowledge representation, Bayesian inference, statistical methods of machine learning, and connectionist ideas about brain-inspired neural networks, which have given rise to the contemporary developments of deep learning. We present a range of real-world applications of AI; in particular, the course will present a catalogue characterising AI research in Aotearoa/NZ as well as technological background towards realising ethical AI. We examine both theoretical and practical components and the content is designed to be accessible to students that do not have extensive programming experience. This course is only available for MAI, PGDipAI, PGCert AI.

Course Learning Outcomes

CLO #	Outcome	Programme Capability Link
1	Identify and explain a range of fundamental approaches and techniques at the core of AI.	
2	Examine some milestone AI systems and their historical role in the development of AI.	
3	Apply and explain symbolic AI and knowledge representation methods in problem solving.	
4	Identify and explain ML methods in classification and regression tasks.	
5	Critically explain and discuss how different AI methods can be combined to form AI systems.	
6	Outline and explain how AI techniques apply to the New Zealand context.	

Additional Information on Assessment

A pass mark in both the practical (Online quizzes, Group work and Presentation) and theory (Test and Exam) components of the course is required to pass the course.

The Group work component is a litterature review in which teams of students have to explore the academic landscape of an AI topic of their choice. It is assessed through a final group report as well as intermediate deliverables.

Assessment Summary

#	Type	%	Classification
1	Group work	40%	Group Coursework
2	Bi-weekly online quizzes	10%	Individual Coursework
3	Presentation	10%	Individual Coursework

#	Type	%	Classification
4	Mid-semester Test and Final Exam	40%	Individual Examination

Assessment Type to Learning Outcome Map

	1	3	4	5	6	8
Group work	●	●		●	●	●
Bi-weekly online quizzes	●	●	●	●		●
Presentation	●	●		●	●	●
Mid-semester Test and Final Exam	●	●	●	●	●	●

Workload Expectations

This course is a standard 15 point course and students are expected to spend 10 hours per week involved in each 15 point course that they are enrolled in.

For this course, you can expect 36 hours of lectures, 60 hours of reading and thinking about the content and 54 hours of work on assignments and/or test preparation.

Delivery Mode Details

Attendance is expected at scheduled activities including lectures to complete components of the course. Lectures will be available as recordings. Other learning activities including seminars will not be available as recordings.
Attendance on campus is required for the exam.
The activities for the course are scheduled as a standard weekly timetable.

Key Topics

The topics covered in the course are likely to change slightly from one delivery to another, to follow trends in AI developments. Some of the key topics covered in S1 2024 were:

- Symbolic Logic
- Reinforcement Learning

- Markov Chain
- Rule base
- LLM based FunSearch
- Expert Systems
- Decision Trees
- Soft Computing
- Genetic Algorithms
- Embodied AI
- Transformer Model + DNN
- Self-supervised learning
- Continual learning
- Time Series/Data Streams

Special Requirements

Attendance is expected at scheduled activities including lectures to complete components of the course.

Learning Resources

Taught courses use a learning and collaboration tool called Canvas to provide students with learning materials including reading lists and lecture recordings (where available). Please remember that the recording of any class on a personal device requires the permission of the instructor.

Additional Information on Learning Resources

List of readings is available on Canvas and Talis.

Disclaimer

Elements of this outline may be subject to change. The latest information about taught courses is made available to enrolled students in Canvas.

Students may be asked to submit assessments digitally. The University reserves the right to conduct scheduled tests and examinations online or through the use of computers or other electronic devices. Where tests or examinations are conducted online remote invigilation arrangements may be used. In exceptional circumstances changes to elements of this course may be necessary at short notice. Students enrolled in this course will be informed of any such changes and the reasons for them, as soon as possible, through Canvas.